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Inner defect detection of glass fiber reinforced polymer sandwich panel using pulsed terahertz imaging based on smoothing and derivative

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ABSTRACT

We conducted an experimental investigation to localize and detect internal defects in a sandwich panel made of glass-fiber-reinforced polymer (GFRP) using terahertz (THz) time-domain spectroscopy and signal processing techniques, which included smoothing and derivative analysis to enhance the defect detection performance. A simulated defect was created using a peel ply and scanned using a pulsed THz scanner with a reflection geometry specifically designed for this purpose. The reflected THz wave was analyzed to improve defect detection performance through time-domain signal processing using smoothing and derivative analysis. To quantify the performance of the proposed signal processing method, we introduced the Defect to Noise Ratio and evaluated its effectiveness accordingly. As a result, noise was removed and defect detection performance was enhanced, enabling the visualization of foreign insert defects in the GFRP sandwich panel.

1. Introduction

A glass-fiber-based sandwich panel is manufactured by bonding a glass-fiber-reinforced polymer (GFRP) plate to a low-density honeycomb or urethane foam. GFRP structure secures the required performance by changing the facial sheet and core materials alongside the structures [1,2]. This composite material can replace metal materials owing to its advantages in terms of weight reduction and mechanical strength. The demand for glass-fiber-based sandwich panels in the aerospace, wind power, construction, and leisure industries are increasing because they have excellent impact resistance and are suitable for structures that require high flexural rigidity. However, these panels may have various defects, such as delamination, voids, de-adhesion, cracks, and hardening over their life cycle.

Many research groups have investigated and developed NDT & E [3–15]. These include ultrasound testing (UT) [7,8], magnetic particle testing (MT) [9,10], electromagnetic testing (ET) [11], thermal infrared testing (IR) [12,13], and Shearography [14,15], each of which has certain advantages and disadvantages. MT can only detect surface cracks. IR requires an external thermal source to improve performance; however, the material of the structure undergoes thermal deformation when exposed to heat sources over an extended period. UT, which is a general method, requires a liquid couplant to efficiently transmit ultrasound signals to the specimen, whereas air-coupled UT does not have this limitation. However, there are problems such as: (1) penetration depth limitation in the reflection mode, (2) image resolution limitation owing to the focused beam size, and (3) signal attenuation in air.

Terahertz (THz) time-domain spectroscopy (TDS) combines several unique characteristics. It has better penetration depth than infrared radiation and better spatial resolution than microwave radiation. Hence, THz technology is well-suited for various imaging applications [16,17]. THz imaging is a powerful tool in NDT & E due to its ability to penetrate non-conductive materials such as plastics, ceramics, composites, and paper, while also providing high-resolution imaging. Here are some advantages of THz imaging in NDT: 1) Non-destructive testing: THz imaging can penetrate materials without causing damage, making it a valuable technique in non-destructive testing; 2) High resolution: THz imaging can provide high-resolution images of materials, revealing details that may not be visible with other imaging techniques; 3) Versatility: THz imaging can be used to detect defects, cracks, and voids in a variety of materials, including ceramics, composites, and plastics; 4) Safety: THz imaging uses non-ionizing radiation, making it a safe technique for operators and objects being imaged. 5) Sensitivity: THz imaging can detect subtle changes in materials, making it a useful tool for detecting defects and monitoring changes over time.

Overall, THz imaging is a promising tool for non-destructive testing due to its ability to provide high-resolution images, detect defects and voids in a range of materials, and do so safely and quickly.

In recent years, several studies on the defect detection of sandwich composites, including GFRP, using THz spectroscopy technology have

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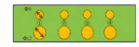
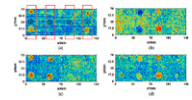
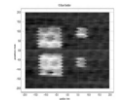
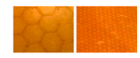
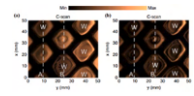
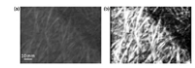
Table 1
Summary of existing defect visualization method using THz wave.

Authors	Sample Info.		Imaging Method	Defect Types	Image	
	Skin (Thickness)	Core			Defect	Results
Wang et al. [18]	GFRP (1.5 mm)	Polypropylene (18 mm)	Freq. Domain	De-bonding (PTFE film)		
Im et al. [19]	GFRP (5.79 mm)	Honeycomb (20 mm)	Time domain	Foreign material (Brass foil)		
Mieloszyk et al. [20]	GFRP	Aramid Polymer	Freq. domain	Foreign material (water)		
Zhang et al. [21]	GFRP (0.5 mm)	Polypropylene (18 mm)	Freq. domain	De-bonding		
Matheis et al. [22]	GFRP (1 mm)	Honeycomb (5 mm)	Freq. domain	Air gap & De-bonding (Teflon sheet)		

been reported [18–22]. Table 1 summarizes several of the proposed THz-based visualization methods. Wang et al. [18] demonstrated that de-bonding content can be successfully extracted from a GFRP skin sandwich panel using the frequency amplitude imaging method. Im et al. [19] applied reflection methods using a THz-TDS system to detect foreign materials fabricated using brass foil. Mieloszyk et al. [20] found that empty and full cells in a metal honeycomb filler could be easily distinguished owing to the different absorption coefficients of electromagnetic waves in the THz range for air and water. Furthermore, Zhang et al. [21] investigated the efficacy of THz-TDS imaging technology for detecting hidden defects in aircraft glass fiber sandwich composites. Matheis et al. [22] inspected various defects, such as air gaps, delamination, and foreign matter, using a terahertz frequency-modulated continuous wave (FMCW) imaging system.

Defect images, such as impact and artificial delamination, have been obtained using conventional THz imaging methods in both the time domain (slice image, peak-to-peak, and magnitude map) and frequency domain (amplitude and phase method). However, this method may not be suitable in situations where it is difficult to determine the signal-to-noise ratio (SNR) and derive a significant difference, such as in substances with small differences in refractive index or subjects with severe attenuation. Although various defect detection techniques proposed in previous studies have yielded good results in well-prepared environments (experimental setup, specimens, etc.), these problems must be overcome for their application in real structures.

In this study, the following steps were taken: 1) A specimen was fabricated using a peel ply impregnated with an epoxy resin; 2) The THz signal acquired at room temperature in a chamber-free environment was analyzed and noise was removed using a signal processing method involving smoothing and derivative analysis to detect defects; 3) We aim to improve defect imaging performance based on the suggested signal processing method.

2. Fabrication of the glass fiber-based sandwich panel with defect

The pulsed THz imaging technique detects defects using physical changes, such as the size or phase delay of the THz waveform measured by the transmission or reflection method. Although visualization is often performed after the fast Fourier transformation of time-series data in a frequency domain, securing the SNR of time-series data should be prioritized. When a pulsed THz wave is incident in the reflection mode, it is difficult to measure a normal signal owing to the damping and incident angle when the thickness of the object is 5 mm or more. If the

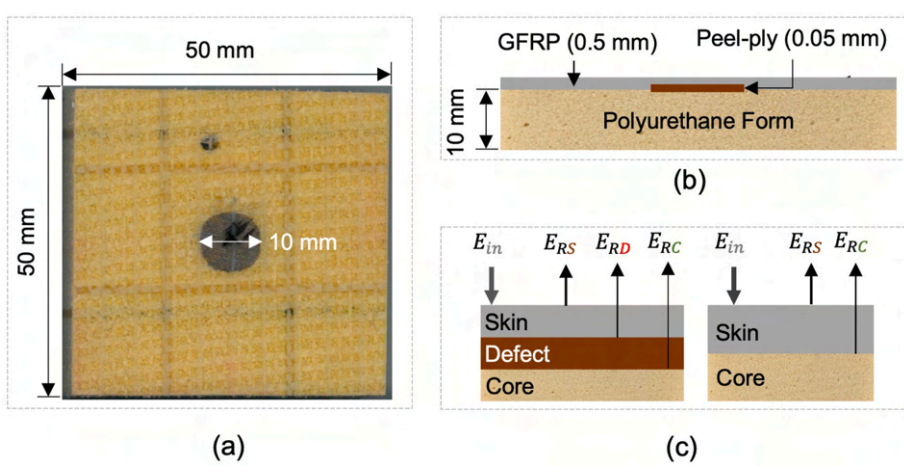


Fig. 1. (a) Optical image of the glass fiber honeycomb sandwich panel; (b) Simulated defect information; (c) Schematic of the Incidence and reflection THz wave at interface.

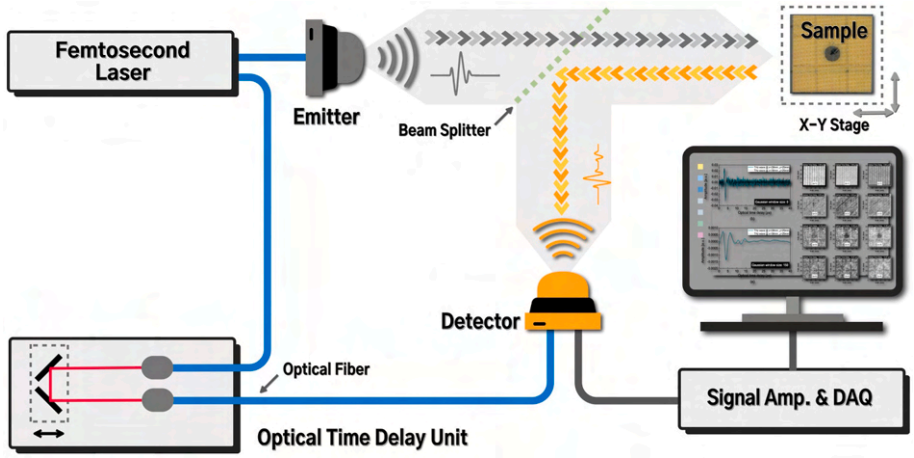


Fig. 2. Schematic test setup for raster scanning with reflection geometry.

difference in components of the specimen is small, it is difficult to distinguish signals reflected at the boundary because of the similar reflective index values, and the larger the noise component, the lower the probability of defect detection and more difficult it is to predict the shape, location, and quantitative size of the specimen.

The pulsed THz imaging technique detects defects using physical changes, such as changes in size or phase delay of the THz waveform measured through the transmission or reflection method. Although visualization is often performed after fast Fourier transformation of time-series data in the frequency domain, prioritizing the SNR of time-series data is crucial. In reflection mode, it can be difficult to measure a normal signal when the object's thickness is 5 mm or more due to damping and incident angle. When the difference in specimen components is small, it can also be challenging to distinguish signals reflected at the boundary due to similar reflective index values. Additionally, higher noise components lower the probability of defect detection and

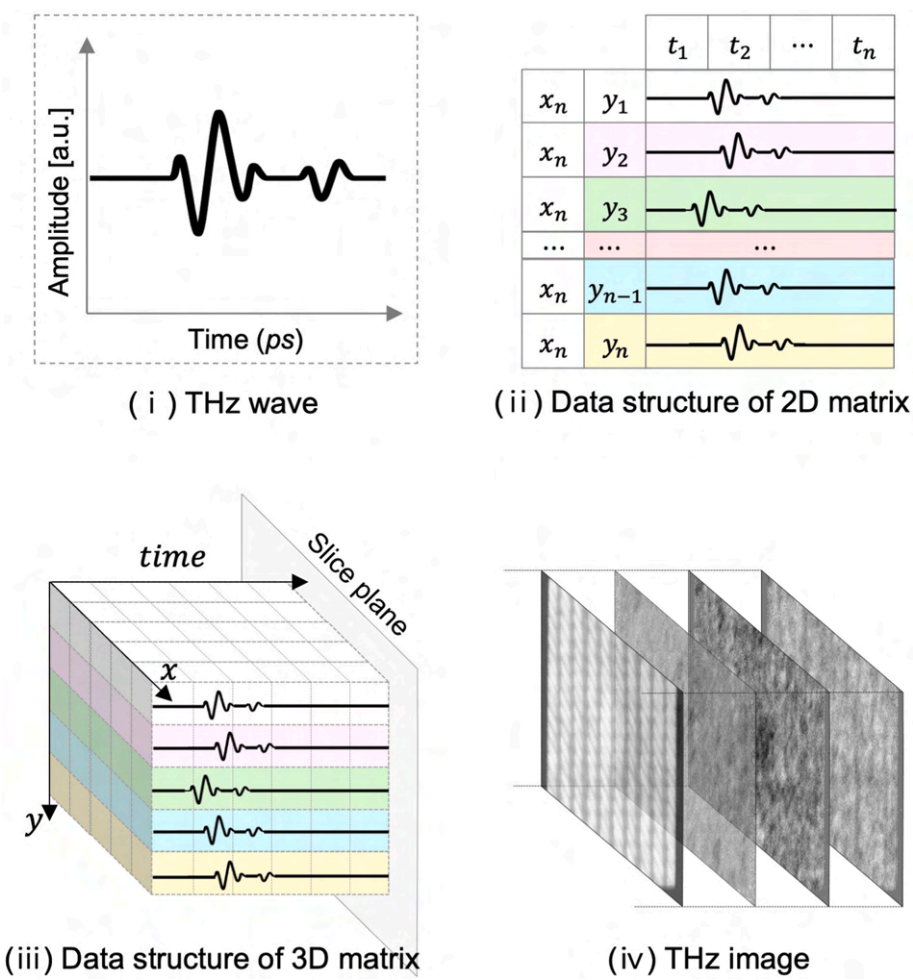


Fig. 3. THz image generation process based on raster scanning.

make it more difficult to predict the specimen's shape, location, and quantitative size.

In this study, epoxy resins were impregnated with glass fibers and peel ply to produce specimens with media characteristics similar to those of defective areas, verifying the defect detection performance. As shown in Fig. 1(a), the specimen had a sandwich structure of GFRP and polyurethane foam, with a peel ply inserted between them. Fig. 1(b) shows the internal defect manufactured by inserting a peel ply with a thickness of 0.05 mm and a diameter of 10 mm between one sheet of unidirectional (UD) glass fiber with a thickness of 0.5 mm and a polyurethane foam core, followed by impregnation and hardening of the epoxy resin. The dark brown grid shape is a slit processed in the urethane foam to improve adhesion with GFRP. Fig. 1(c) displays the pulsed THz waves reflected at the plane of incidence and the interface in the normal and defective areas. The pulsed THz wave is denoted by *E*, and the incident reflected signal is distinguished by the first suffix. The THz wave irradiated on the inspection object is defined as in, the reflected THz wave as *R*, and the interface of each medium is distinguished by the second suffix. The GFRP skin, reflected signal from the defect, and reflected signal from the core are defined as *S*, *D*, and *C*, respectively.

3. Reflection-based pulsed THz imaging system

We utilized a commercially available fiber-coupled THz-TDS system (TERA K15, Menlo Systems Corp.) to detect and visualize the internal defects in the GFRP sandwich panel. The test setup for defect detection was similar to that used in a previous study [17]. However, we aimed to improve the defect detection performance through enhanced signal processing methods suggested in this study. The THz-TDS system had a bandwidth ranging from 0.1 to 4 THz, with a guaranteed dynamic range greater than 75 dB and a total scanning range greater than 850 ps. In this study, the scanning range was 80 ps, and the THz-TDS system provided an average THz-wave power output of over 60 mW. The TERA K15 system consisted of a femtosecond laser (T-Light, Menlo Systems Corp.), an optical delay unit (ODU, Menlo Systems Corp.), PCAs for the emitter (TERA 15-TX-FC, Menlo Systems Corp.), detector (TERA 15-RX-FC, Menlo Systems Corp.), and a two-axis motorized linear stage (Precision linear stage, Physik Instrumente GmbH & Co Corp.), as shown in Fig. 2. The emitter and detector were positioned at a 90-degree angle using a beam splitter, and the THz wave was incident on the specimen surface at a 90-degree angle.

After securing the test specimen on the x-y axis motorized stage, the raster scan method was used to perform the test, and THz waves were collected as time-series data at each location. The delay length of the THz wave measured at one point using the THz-TDS system was 80 ps. The defect visualization technique used in the previous study was employed in both the time and frequency domains. The visualization process in the time domain using a captured THz waveform is known as THz pulse time-domain visualization (TTV), while that in the frequency domain is called THz pulse frequency-domain visualization. The visualization process, depicted in Fig. 3, involves the following steps: (1) Measuring the THz signal at each pixel; (2) During THz signal measurement, the motorized stage affixed to the sample moves in the x direction, then in the y direction when the end point is reached, before moving to the start point and continuing with the measurement; (3) Reshaping the scanned THz waves into a three-dimensional (3D) matrix (x-axis, y-axis, time); and (4) Slicing the plane according to time or frequency and subsequently imaging each signal in 2D [17].

4. SNR enhancement method based on smoothing & derivative for THz imaging

To ensure the THz wave penetrates the specimen, it must be incident perpendicular to the surface. To achieve this, 46% of the total THz waves emitted from the generator are reflected in free space, while the remaining 54% (E_{in}) is irradiated into the specimen via the beam splitter. The THz waves reflected from the specimen surface (E_R) return to the beam splitter, with 46% of them penetrating the specimen and reaching the detector, and the remaining 54% radiating into free space. However, only up to 24.8% of the energy radiated from the generator is effective for the THz wave E_R acquired from the detector, resulting in a lower SNR and amplitude. Moreover, when the signal is measured after penetrating a specimen with a thickness of several millimeters or more, the SNR is further reduced due to damping inside the specimen, making it difficult to distinguish between a normal signal and a defect signal.

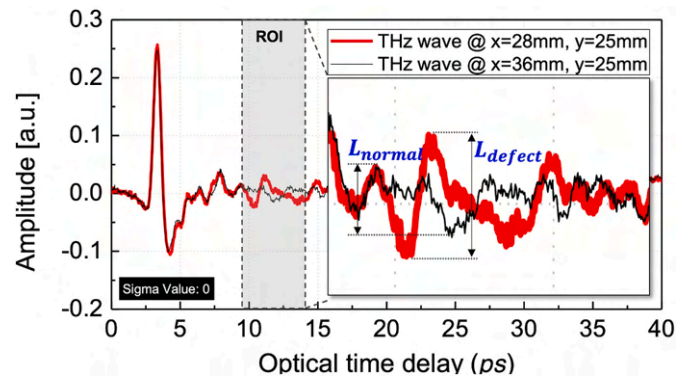


Fig. 4. Definition of the L_{normal} and L_{defect} in ROI of the THz wave.

The measured signal also includes various types of electrical noises. These noises originate from the 100 VDC voltage applied to the emitter to generate THz, the current detector connected to the detector, the data acquisition system (DAQ) that converts analog signals into digital signals, and other electrical equipment. When the SNR is high, the defect signal can be distinguished clearly without removing the noise signal. However, when the SNR is low, it becomes difficult to distinguish between the defect and noise signals.

In this study, we aim to evaluate the performance enhancement of THz imaging in the region of interest (ROI) by measuring the modified SNR. The modified SNR takes into account the improvement in signal-to-noise ratio achieved by taking the derivative of the signal, and is referred to as the Defect-to-Noise Ratio (DNR). DNR is calculated as the ratio of the defect signal level to the normal signal level using the following equation (1):

$$DNR = L_{defect} \,/\, L_{normal} = 20\log\left(L_{defect} \,/\, L_{normal}\right)\ [dB] \tag{1}$$

where L_{defect} represents the signal level in ROI of the THz wave, while L_{normal} is the noise level in the same ROI of the THz wave as illustrated in Fig. 4.

4.1. Smoothing effect on the pulsed THz wave with noise component

One common method for removing noise signals is to use an averaging technique, in which signals are measured multiple times at the same point and then averaged [23]. The normal signals obtained are not distorted since the repeatedly measured signals have the same characteristics. However, noise signals tend to converge to zero due to the lack of cycle and regularity. Since pulsed THz waves are acquired using an ODU, it takes a significant amount of time to acquire signals, and therefore, the averaging technique is not suitable. An alternative to the averaging technique, which can acquire THz waves and remove noise, is the smoothing technique [24]. Various filters can be used for smoothing, and in this study, a Gaussian filter was used [25]. This is because it is an ideal probability model that best expresses natural phenomena and is suitable for removing noise signals. The degree of smoothing in Gaussian filter-based noise removal is determined by the sigma size (σ), which represents the window size. A larger sigma size improves noise removal performance, but increases signal distortion.

For comparison of THz waves using different sigma sizes (σ), sigma

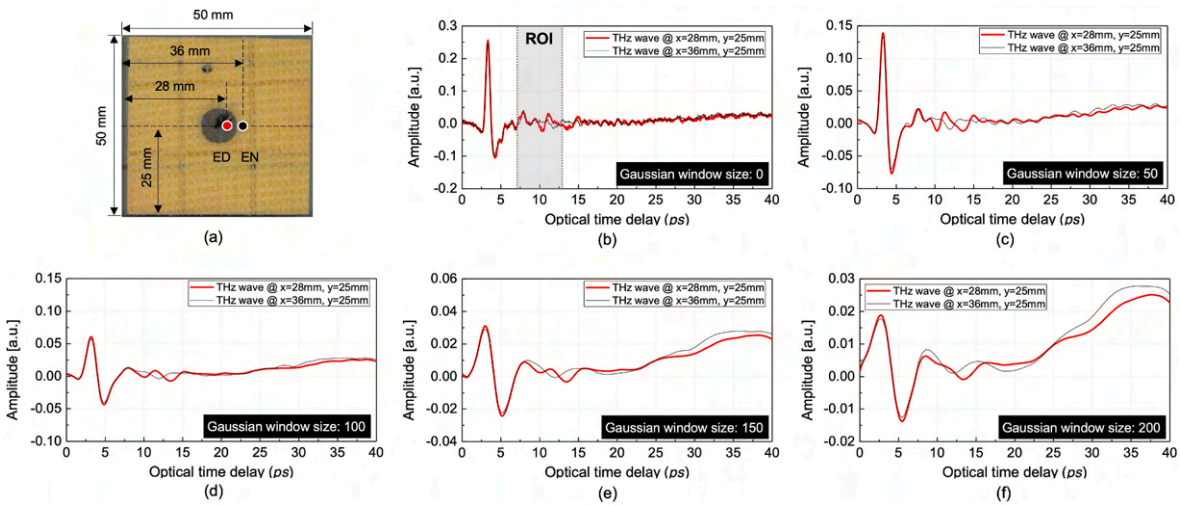


Fig. 5. (a) Representative THz wave locations map; Smoothing effect results according to the Gaussian window size: (b) $\sigma = 0$; (c) $\sigma = 50$; (d) $\sigma = 100$; (e) $\sigma = 150$; (f) $\sigma = 200$.

sizes of 50, 100, 150, and 200 were used, and the results are summarized in Fig. 5. Fig. 5(a) shows the location of the signal selected for signal analysis. The defect is located where the peel ply is inserted, at x = 28 mm and y = 25 mm, while the normal area is at x = 36 mm and y = 25 mm, which consists only of glass fiber and urethane foam. Fig. 5(b) presents the normal and defect signals that were not processed. The thick solid line represents the defect signal, and the thin solid line represents the normal signal. As the sigma size increases from 50 to 200, the noise signal decreases, but the signal distortion increases, making it difficult to distinguish between normal and defective signals. It was determined in this study that a sigma size of up to 150 can be applied when smoothing; however, it is necessary to adjust the threshold to suit the type of test specimen and test environment. This is because the maximum value of the THz waves after 20 ps is larger than the size reflected from the first surface E_{R1}, as shown in Fig. 5(f). Furthermore, a distinguishable reflective signal is distorted at approximately 12 ps, making it difficult to observe a peak corresponding to a defect.

The results of the visualization using these characteristics are discussed in detail in Section 4.2. It was found that the smoothing technique, applied to remove the noisy signal, shows improved performance at certain sigma sizes. However, constructing 3D shape visualization and projection defect images inside specimens requires a signal processing process that can improve the SNR without distorting the signal.

4.2. Derivative effect on the pulsed THz wave for improving the SNR

When the defect size is small, interlayer separation at the micrometer level or insignificant differences in material properties can cause the THz signal corresponding to the defect location to be buried under multiple noise signals, making it difficult to detect. Although a smoothing technique was applied to address this issue, the distortion of the defect signals hidden at 10–15 ps was severe, and abnormal signals increased after 20 ps.

In spectroscopy using ultraviolet or near-ultraviolet rays, it is often challenging to accurately distinguish materials due to absorption area overlapping or interference. To solve this problem, derivative spectrophotometry has been used to quantify the rate of change in absorbance of each material by identifying wavelengths at which several mixtures do not affect each other [26]. While derivative spectrophotometry cannot be directly applied to THz imaging, the discontinuity of THz waves reflected from the boundary of the defect can be calculated. This method is similar to the mode shape curvature technique that extracts the mode shape of a defective structure in a noisy vibration field [27]. Differential-based signal processing was used to determine the point at which the extreme value changed in the measured waveform. The point passing through the origin becomes an extreme value, and the negative and positive zero-crossing points indicate the position information of the

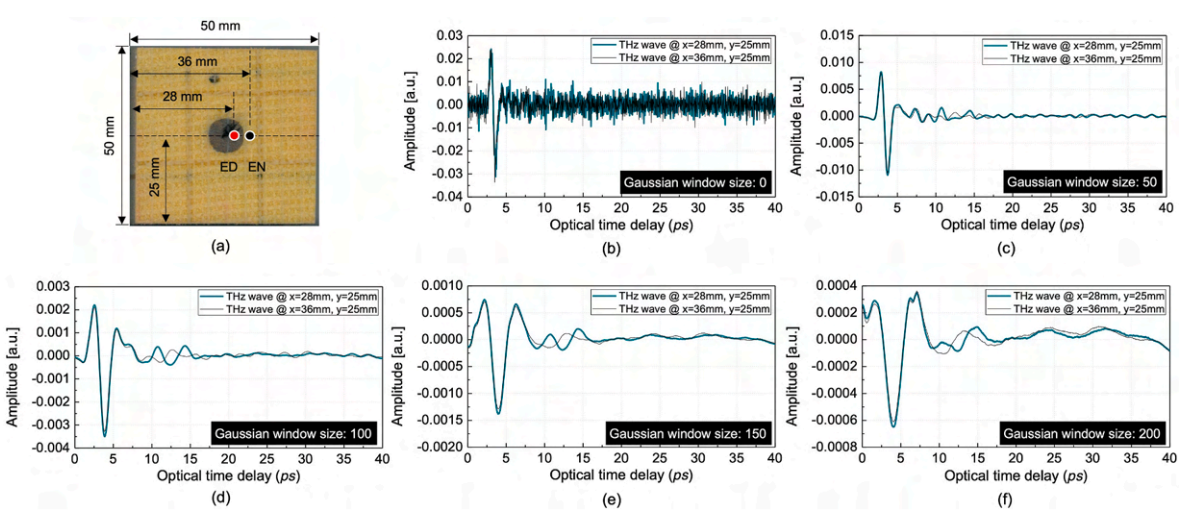


Fig. 6. (a) Representative THz wave locations map; Derivative effect after smoothing process: (b) $\sigma = 0$; (c) $\sigma = 50$; (d) $\sigma = 100$; (e) $\sigma = 150$; (f) $\sigma = 200$.

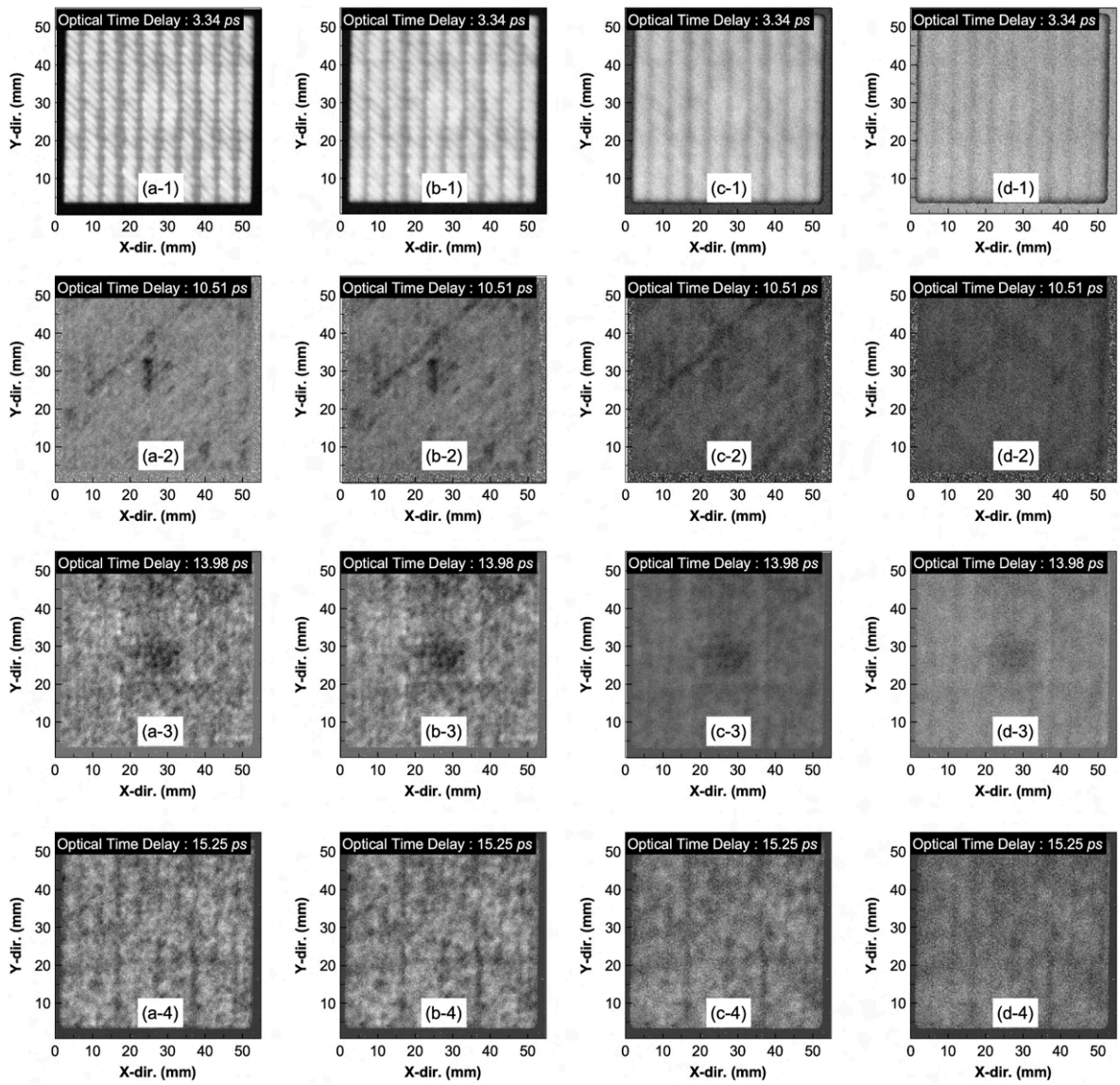


Fig. 7. S-TTV C-Scan results according to the optical time delay: (a) w/o smoothing process; (b) sigma size = 50; (c) sigma size = 100; (d) sigma size = 150.

peak with the maximum and minimum values, respectively [28]. Therefore, it is possible to identify the discontinuity of the signal with ease.

Fig. 6 (b) presents the results of differentiating THz waves without signal processing. The derivative amplifies any noise signal present, which negatively impacts the SNR. The change in the extreme value of the noise signal is greater than that of the defect signal, making it difficult to secure the SNR. However, if the derivative is applied after removing the noise signal through smoothing, the SNR can be secured in the area where the defect signal exists. Fig. 6 (c)–(f) illustrate the ability to eliminate the phenomenon where the magnitude gradually increases after 20 ps. Fig. 6 (c) shows the results of differentiating the smoothed signals presented in Fig. 5 (c). Unlike the results of Section 4.1, where distinguishing between normal and defective signals was impossible, a significant difference between the two signals was confirmed after approximately 10 ps. The results obtained are presented more clearly in Fig. 6 (c)–(e). However, when the smoothing value was high ($\sigma = 200$), as shown in Fig. 5 (f), the original signal was distorted, indicating the need for appropriate selection of the smoothing value.

5. Defect visualization results based on smoothing and derivative process

A peel ply, impregnated with epoxy resin and with a thickness of 0.05 mm, is positioned between the glass fiber and urethane foam. It exhibits material properties very similar to GFRP in terms of reflective index, meaning that the THz wave energy reflected by the surface at the interface between the GFRP and peel ply is very low. The visualization images obtained using the existing TTV are presented in Fig. 7 (a1) – (a4). With the increase in optical time delay, a cross-sectional image in the depth direction from the specimen's surface is displayed. Fig. 7 (a2) – (a3) show the periphery of the interface between the GFRP and peel ply, where it is possible to identify a dark area that corresponds to the peel ply. However, it is challenging to determine this area as defective because its size and shape are unclear.

5.1. Defect image visualization results with only smoothing process

Fig. 7 (b)–(d) show the application of a noise removal method, the smoothing-based THz waves, to the TTV visualization technique to improve the SNR and remove noise. The Gaussian filter was used to smooth the entire time-series data of THz waves and the A-scan results

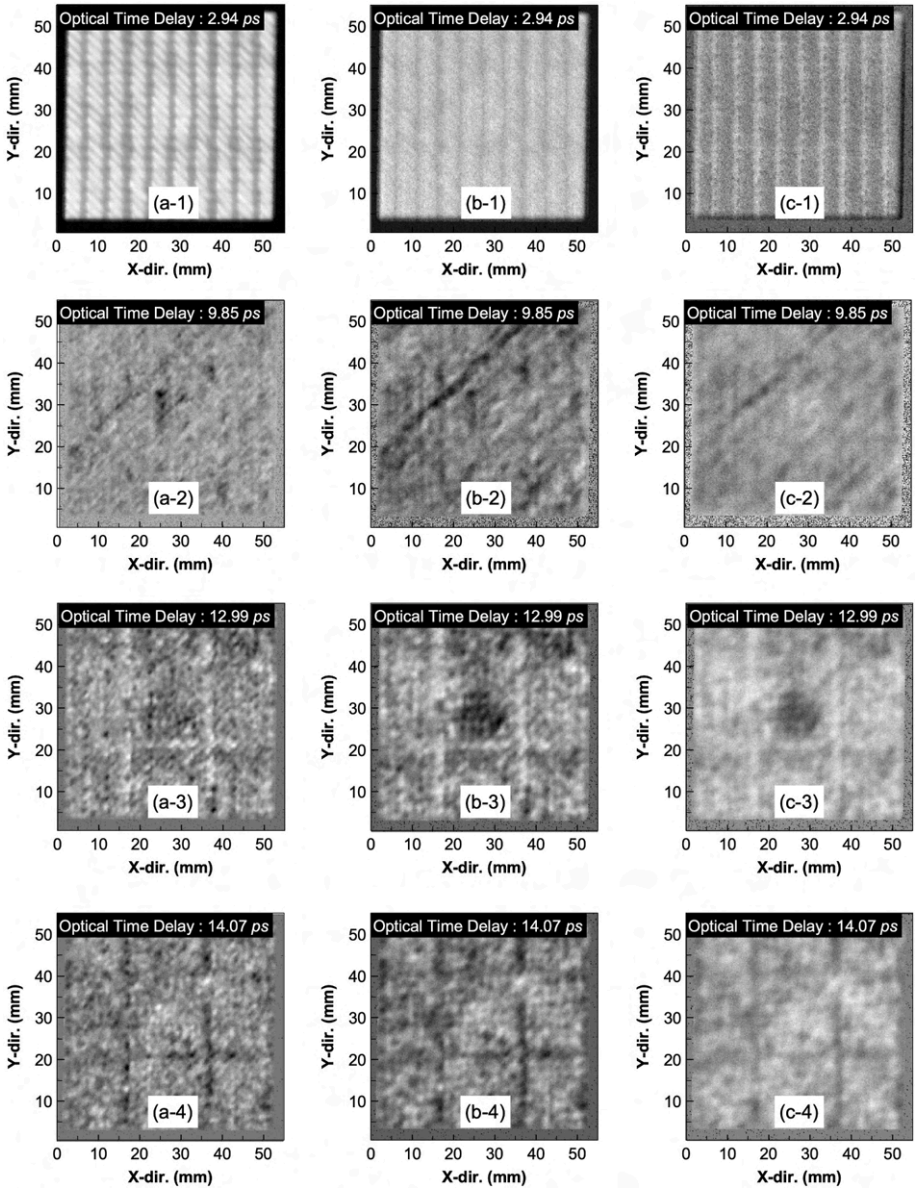


Fig. 8. SD-TTV C-Scan results according to the optical time delay: (a) sigma size = 50; (b) sigma size = 100; (c) sigma size = 150.

that make up the image pixels. Four sigma sizes were tested: 50, 100, 150, and 200. However, as shown in Fig. 7 (d1) - (d4), the signal distortion was greater than the peak reflected from the specimen surface when the sigma size was set to 200. Therefore, visualization performance was evaluated and compared for three window sizes: 50, 100, and 150.

The results obtained with Gaussian window sizes of 50, 100, and 150 are shown in Fig. 6 (b)–(d), respectively. The TTV technique with only smoothing applied is defined as S-TTV. Although it was possible to identify the circular defect located at the center when the Gaussian window size was more than 100, the grid-shaped slit, fiber direction of the GFRP skin, and diagonal shapes had increased signal distortion, resulting in reduced visualization performance. The next section evaluates the effect on the differential, and the sigma size is set to 50, 100, and 150.

5.2. Enhanced defect image visualization results with derivative after smoothing process

To address the issues encountered in Section 5.1 with S-TTV, the THz waves corresponding to each pixel in the image were smoothed and then differentiated. This technique, referred to as SD (smoothing Derivative)-TTV, successfully visualized the defective image. Fig. 8 (a1) - (a4), Fig. 8 (b1) - (b4), and Fig. 8 (c1) - (c4) show the results of performing the derivative with smoothing sizes of 50, 100, and 150, respectively. Visualization using SD-TTV was conducted at depths of 9.85, 12.99, and 14.07 ps from the surface of the specimen at 2.94 ps. As shown in Fig. 8 (a1) - (c1), it was possible to identify the fiber direction after additional derivative following smoothing, even though the direction was blurred at the specimen surface as shown in Fig. 7 (b1) - (d1). Additionally, Fig. 8 (c3) clearly shows a peel ply with a diameter of 10 mm located in the center of the specimen, even with a smoothing size of 150. Moreover, Fig. 8 (c4) displays a visualization of the grid-shaped slit shape, which was not observed in the S-TTV results.

Previous studies proposing TTV [17] have shown improved visualization performance compared to techniques using peak-to-peak size and delay values, which are commonly used in THz imaging. However, there are limitations to the visualization performance. To address this, this study applied smoothing and derivative simultaneously to THz imaging, which are techniques used in the noise vibration and spectroscopy fields,

Table 2
Summary of DNR values according to smoothing size after derivative.

Sigma size	Defect Signal [a.u.]		Norma Signal [a.u.]		DNR [dB]
	Max	Min	Max	Min	
50	0.0017	−0.0021	0.0013	−0.00208	1.01
100	0.00042	−0.00041	0.00025	−0.00032	3.57
150	0.000213	−0.000181	0.000153	−0.000151	2.16
200	0.000114	−0.000075	0.000071	−0.000112	0.19

and confirmed their applicability. Furthermore, this study demonstrated that SD-TTV, a signal-processing method suitable for THz imaging, improved the visualization performance.

Smoothing and derivative processing are useful signal processing methods for extracting defect information. Table 2 summarizes the results of these methods using DNR. The highest DNR value of 3.57 dB was observed at a sigma size of 100, while for sigma sizes of 50 and 200, DNR values of 1.01 dB and 0.19 dB were calculated, respectively. This indicates that the defect detection capability and DNR values exhibit the same tendency as seen in the A-scan results that can be visually confirmed by humans and the corresponding defect images.

6. Conclusions

In this study, we detected internal defects in a glass fiber-based sandwich panel using pulsed THz waves that penetrate non-metallic materials and reflect from material-changing interfaces. We proposed a signal-processing method that included smoothing and derivative signal processes to improve defect detection performance under harsh inspection conditions. To achieve this, a peel ply with a thickness of 0.05 mm, 1/10 of the glass fiber thickness, was inserted between the glass fiber and urethane foam.

The inspection was conducted in reflection mode based on a raster scan, and noise removal (smoothing) was performed to extract the signal (ERP) reflected from the interface between the glass fiber impregnated with epoxy and the peel ply. The performance of the smoothing and derivative signal processing was evaluated using DNR and THz imaging. Although the C-scan image suggested in the previous study had limitations in visualizing peel ply and slit shapes simulated in the specimen, the suggested SD-TTV was able to successfully visualize all orientations, defects, and slits of the GFRP sandwich panel, enabling accurate identification of defect location and shape without requiring professional knowledge.

In the future, the SD-based signal-processing technique can be further expanded to improve the accuracy of thickness measurements for thin films, such as coatings and paintings, and to improve prediction accuracy based on machine learning. We also plan to include sandwich panels according to the resin curing conditions and develop various methods to predict the properties of composite structures, aiming to improve the level of technology.

Authorship statement

Category 1: Conception and design of study: **D.-H. Han**
acquisition of data: **D.-H. Han**
analysis and/or interpretation of data: **D.-H. Han**.
Category 2: Drafting the manuscript: **D.-H. Han**
revising the manuscript critically for important intellectual content: **D.-H. Han**.
Category 3: Approval of the version of the manuscript to be published (the names of all authors must be listed): **D.-H. Han**.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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